Evaluation Choices Determine Apparent Quantum Kernel Robustness under Distribution Shift

Roberto Fernández-Barrios[ORCID][1*], Iker Pastor-López[ORCID][1], Asier González-Santocildes[ORCID][1], Pablo García Bringas[ORCID][1]

[1*]Faculty of Engineering, University of Deusto, Avda. de las Universidades, 24., Bilbao, 48007, Spain.

*Corresponding author(s). E-mail(s): roberto.fernandez.b@deusto.es; Contributing authors: iker.pastor@deusto.es; gonzalez.asier@deusto.es; pablo.garcia.bringas@deusto.es;

Abstract

Quantum-kernel robustness claims depend on how candidates are tuned and selected. We audit this dependence with a controlled kernel-swap benchmark, a ten-specification curve, and a prospectively frozen external validation on three real-world domain shifts. In security benchmarks, fixed regularization, weak references and same-test oracle selection produce apparent gains up to $\mathbf{+0.037}$, whereas train-only tuning, target-label-free selection and equal budgets yield quantum-minus-classical effects of $\mathbf{-0.005}$ for support-vector classification and $\mathbf{-0.002}$ for Gaussian-process classification. Externally, the controlled support-vector effect is $\mathbf{-0.0119}$ (conditional 95% interval $[\mathbf{-0.0205, -0.0033}]$), while the weak-oracle effect is $\mathbf{+0.0089}$ and protocol contraction is $\mathbf{+0.0208}$. Contraction also remains positive for the probabilistic classifier, although its controlled effect crosses zero. Geometry is associated with accuracy but shared by both families, and finite-shot estimation distorts exact-statevector rank. The apparent robustness advantage is therefore an evaluation-dependent result in the low-qubit regimes studied.

Keywords: quantum machine learning; quantum kernels; distribution shift; out-of-distribution robustness; kernel methods; kernel-target alignment; Gaussian processes; malware detection; network intrusion detection

1 Introduction

Quantum kernel methods embed classical data into quantum states and use state overlaps as similarities for otherwise classical learners Havlíček et al. (2019); Schuld and Killoran (2019); Schuld (2021). This separation makes them a clean setting for asking whether quantum-induced feature geometry provides useful inductive bias. It also makes comparison unusually sensitive to evaluation design: the downstream learner can be held fixed, but conclusions still depend on the encoding, bandwidth, regularization, classical reference set, and rule used to select a candidate Huang et al. (2021); Schuld et al. (2021); Bowles et al. (2024); Schnabel and Roth (2025).

Recent benchmarks have accordingly shifted the question from whether an isolated quantum model can perform well to whether a claimed gain survives strong and symmetric controls. Broad QML studies find competitive classical baselines once pipelines are aligned Bowles et al. (2024); quantum-kernel benchmarks show substantial sensitivity to feature map and hyperparameter choice Schnabel and Roth (2025); and a concurrent tabular study using nested cross-validation, spectral analysis, and hardware validation finds no significant quantum win across its tested pairs Kakavand et al. (2026). These studies evaluate predominantly identically distributed train–test settings. They do not address a further deployment question: what happens when the target distribution changes and its labels cannot be used to choose the deployed kernel?

Distribution shift makes model selection part of the estimand. Work on domain generalization and real-world shift has shown that robustness gains can disappear when search budgets, hyperparameters, and selection criteria are standardized Gulrajani and Lopez-Paz (2021); Koh et al. (2021); Yao et al. (2022); Zhou et al. (2023). A selector that inspects the shifted test labels estimates an oracle upper bound, not deployable performance; a family with more candidates has more opportunities to win that selection. Fair quantum–classical evaluation under shift must therefore control not only the classifier and data split, but also regularization, candidate budget, and label access.

Security detection provides a useful stress test because shift is intrinsic to the application. Malware evolves over time, and experimental bias can overstate detector performance Anderson and Roth (2018); Pendlebury et al. (2019); Arp et al. (2022); network traffic changes with infrastructure and attack campaigns Moustafa and Slay (2015); Alsaedi et al. (2020). Quantum classifiers have already been evaluated on intrusion-detection benchmarks, sometimes with reported gains over small classical reference sets on i.i.d. splits Abreu et al. (2024, 2025); Bellante et al. (2025). We use these datasets not to propose a security system, but to expose kernel families to reproducible constructed shifts and to a held-out attack-campaign shift.

The closest robustness-specific concurrent work studies additive input corruption and proposes a spectral phase encoding Herrero Gómez et al. (2026). Our question is complementary: rather than optimize a new encoding, we audit how evaluation choices change a family-level conclusion under constructed and held-out-campaign shifts. The distinguishing axes are no-OOD-label selection, equal candidate budgets, per-configuration train-only regularization, fixed-case inference, and a geometry analysis tied specifically to behaviour after the distribution moves.

We implement a controlled *kernel-swap* benchmark in which preprocessing, splits, classifier, and family-internal selection logic are fixed while only the kernel changes. Four security scenarios from EMBER, UNSW-NB15, and ToN-IoT yield eight scenario-groups across constructed and held-out-campaign shifts. Support-vector and Laplace Gaussian-process classifiers consume the same precomputed Gram matrices. The quantum family contains four fidelity feature-map families with angle-scale sweeps; the classical family extends linear and RBF references with polynomial, Laplacian, and Matérn kernels and symmetric bandwidth freedom. Each configuration's regularization is tuned from training data alone. The primary deployed candidate is selected on a disjoint ID-validation split, and the two extended families receive equal candidate budgets. Fifteen pipeline realizations per setting quantify conditional variability; the benchmarks are treated as fixed case studies rather than as random draws from an undefined population.

The comparison is coupled to a mechanistic audit. We measure effective rank, centered kernel-target alignment, concentration, and geometric difference Roy and Vetterli (2007); Cristianini et al. (2001); Cortes et al. (2012); Huang et al. (2021); use partial association, leave-one-dataset-out prediction, and effective-rank matching to restrict interpretation; and perturb fidelity estimates with finite shots to test whether exact statevector geometry survives measurement.

The evaluation choices change the conclusion. With same-test oracle selection, fixed regularization, and customary linear+RBF baselines, fidelity kernels appear to improve OOD balanced accuracy by as much as $+0.037$. With train-only regularization, no-OOD-label selection, and equal candidate budgets, the dataset-equal quantum-minus-classical delta is -0.005 for SVC and -0.002 for the GP classifier. The primary SVC comparison is classical-favoured or conditionally tied in seven of eight scenario-groups. Effective rank and OOD alignment remain associated with accuracy, but the association is regime-dependent and shared by both families; at matched effective rank, the classical configuration is at least as accurate in every group. Finite-shot estimation inflates effective rank and destabilizes candidate selection.

The contribution is therefore an evaluation result rather than an advantage claim: it identifies which protocol choices create the optimistic quantum-kernel robustness result, supplies a deployable equal-budget comparison under shift, and shows that the observed geometry does not uniquely identify quantum provenance. The scope is deliberately limited to the low-qubit fidelity kernels, datasets, and shift constructions studied here.

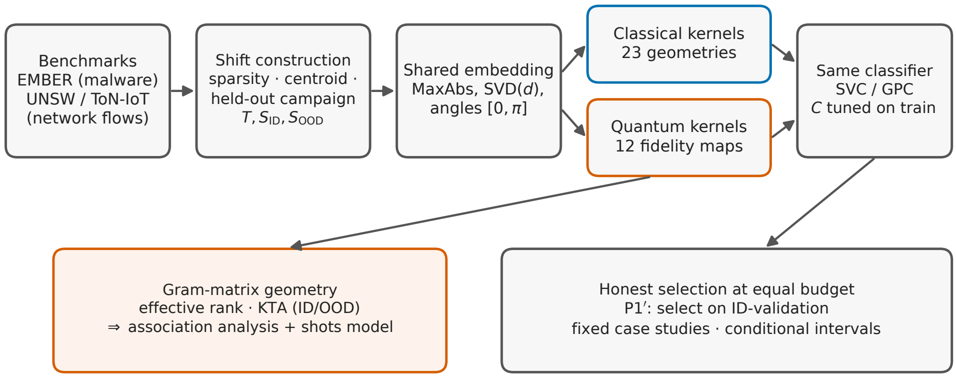

Fig. 1 The controlled kernel-swap protocol. Benchmark pools feed reproducible shift constructions; all kernels operate on a shared train-only embedding, receive per-configuration regularization tuned on the training split, and are consumed by the same precomputed-kernel classifiers. Family-level conclusions are drawn by honest, equal-budget selection on an in-distribution validation split (P1′), summarized as fixed case studies with conditional intervals; the Gram matrices feed the geometry-association analysis and the finite-shot model.

2 Results

2.1 Overview

We report results under the controlled kernel-swap protocol described in Section 4. All comparisons share the same classifier, preprocessing pipeline, and ID/OOD split definitions within a scenario-group; the only varying component is the kernel, and every number below aggregates fifteen pipeline realizations per setting. The argument proceeds in three movements. First, under test-peeking oracle selection and fixed regularization, the fidelity kernels show an apparent OOD advantage over linear and RBF baselines (Section 2.2). Second, we tune each configuration on training data alone, select on an in-distribution validation split disjoint from both test splits, and match the candidate budgets of the extended families. Under this controlled comparison no robust family-level advantage remains; the primary SVC comparison is classical-favoured or tied in seven of eight scenario-groups (Section 2.4). Third, a geometry analysis shows that effective rank and OOD alignment are associated with—but do not causally determine—performance, that the association is regime-dependent, and that it favours neither family (Section 2.6); a finite-shot analysis then shows the exact geometry does not transfer unchanged to finite measurement (Section 2.7). Throughout, we treat the eight scenario-groups as fixed case studies and report effect sizes with conditional intervals rather than a population p-value, because the settings within a benchmark share data pools and are not exchangeable (Section 4).

2.2 The apparent advantage under oracle selection

We begin with the comparison that dominates the literature: fidelity-based quantum kernels against linear and RBF baselines, with each family's representative chosen by Best-by-OOD selection—the configuration with the highest balanced accuracy on the OOD test split itself—and the classical kernels left at the default regularization $C = 1$. Under these conventions, on EMBER, the quantum family shows an OOD advantage of +0.037 under SVC and +0.028 under the GP classifier against linear+RBF, shrinking to +0.004/+0.002 once the classical family is enlarged with heavier-tailed kernels. Supplementary Table 2 gives the complete oracle results.

Two features favour this apparent quantum result. Best-by-OOD is an *oracle*: it uses the labels it then reports. Fixing $C = 1$ also leaves classical candidates at a potentially unsuitable regularization point. These values are therefore descriptive upper bounds, not deployable effects.

2.3 A specification curve locates the reversal

To separate the contribution of each evaluation choice from the choice of a preferred endpoint, we enumerated all ten logically valid, already-computed specifications fixed before this extension (Figure 2). The specifications vary four axes in a prespecified order: same-test OOD-oracle versus ID-based selection, fixed versus train-cross-validated SVC regularization, customary linear/RBF versus extended classical

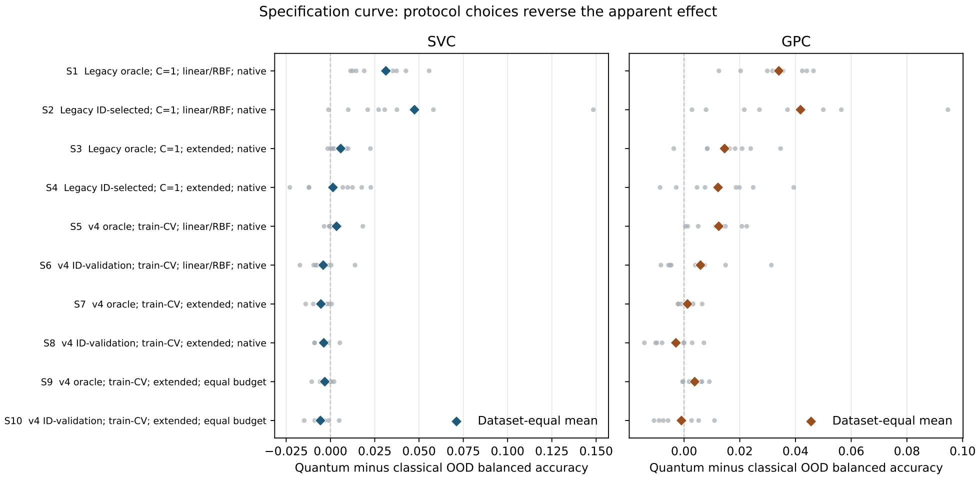

Fig. 2 A frozen specification curve shows that evaluation choices reverse the apparent family effect. Grey points are the eight security scenario-groups; diamonds are dataset-equal means. Specifications are shown in their prespecified S1–S10 order, not sorted by effect. They vary target-label access, SVC regularization, classical reference strength, and candidate budget. Positive values favour the fidelity-kernel family. GPC has no C-regularization axis. No specification-level hypothesis test is performed.

references, and native versus equal candidate budgets. Every specification retains all eight security scenario-groups.

The family-level conclusion spans both sides of zero. For SVC, the dataset-equal quantum-minus-classical mean ranges from $+0.047$ under legacy ID selection with fixed $C = 1$ and the customary reference (S2) to -0.006 under ID-validation selection, train-only regularization, the extended reference, and equal budgets (S10). For the GP classifier, for which the regularization axis is not applicable, the corresponding range is $+0.042$ to -0.003. Expanding the classical reference contracts the positive effect before honest selection or budget matching is imposed; train-only SVC regularization and ID-validation selection then move the controlled estimates to zero or slightly negative. The path is not monotone in every intermediate specification, which is precisely why reporting only an oracle and a final controlled endpoint would be insufficient. The curve shows that the apparent advantage is conditional on a bundle of evaluation decisions rather than a stable property of quantum provenance.

2.4 A fair comparison removes the advantage

A deployable comparison must tune each configuration's regularization without looking at the test data, select the deployed configuration without OOD labels, and compare families at the same candidate budget. We implement all three (Section 4). For every kernel candidate, the SVC regularization C is chosen by five-fold cross-validation on the training Gram matrix alone. The deployed configuration is then the one with the best balanced accuracy on an in-distribution *validation* split, held out from the ID test split (protocol P1′); no OOD label is ever consulted in selection. The primary comparison uses the same candidate budget for both extended families (60 per family in full-coverage groups and 20 in reduced-coverage groups), with the full classical pool retained as a sensitivity. We treat the eight scenario-groups as fixed case studies and report, per group, the quantum-minus-classical OOD delta with a conditional interval over the five split realizations, and no population p-value (Section 4).

Table 1 and Figure 3 give the result: no robust advantage remains against either reference. Against the equal-budget extended classical family, the P1′ delta is classical-favoured or within the conditional uncertainty of zero in seven of the eight scenario-groups under the primary SVC comparison, with a dataset-equal-weighted mean of -0.005 (leave-one-dataset-out range $[-0.006, -0.004]$); under the GP classifier the mean is -0.002 ($[-0.003, +0.000]$). Against the customary linear+RBF baseline—the comparison that produced the $+0.037$ oracle result of Section 2.2—the no-OOD-label delta is -0.004 under SVC and $+0.004$ under the GP classifier, with heterogeneous signs across scenario-groups. The single regime that retains a small quantum lean under both equal-budget comparisons is UNSW-NB15 reconnaissance under held-out attack-campaign shift ($+0.005$ for both classifiers); both conditional intervals include

Table 1 No-OOD-label selection results (eight scenario-groups, fifteen pipeline realizations per setting). Quantum-minus-classical OOD balanced-accuracy delta under P1′ (select on ID-validation, tune per-configuration C on train, evaluate on OOD test). The extended-family comparison is the frozen equal-budget analysis: 60 candidates per family in full-coverage groups and 20 in reduced-coverage groups. Its interval is conditional pipeline-realization uncertainty over the five split seeds. The customary linear+RBF comparison is contextual rather than an equal-budget endpoint; the full-pool extended-family sensitivity remains in the machine-readable artifact. No population p-value is reported.

Scenario	Clf.	vs lin./RBF	vs ext. (equal B)	conditional interval
EMBER $m1$	SVC	−0.0095	−0.0058	[−0.0340, +0.0223]
	GPC	−0.0050	−0.0009	[−0.0101, +0.0083]
EMBER $m2$	SVC	+0.0139	−0.0064	[−0.0137, +0.0009]
	GPC	+0.0149	+0.0027	[−0.0027, +0.0081]
UNSW-DoS (campaign)	SVC	−0.0043	−0.0033	[−0.0139, +0.0073]
	GPC	+0.0041	−0.0057	[−0.0120, +0.0005]
UNSW-DoS (constructed)	SVC	−0.0085	−0.0044	[−0.0119, +0.0031]
	GPC	−0.0056	−0.0108	[−0.0177, −0.0039]
UNSW-Recon (campaign)	SVC	+0.0004	+0.0050	[−0.0058, +0.0157]
	GPC	+0.0074	+0.0052	[−0.0019, +0.0123]
UNSW-Recon (constructed)	SVC	−0.0076	−0.0089	[−0.0146, −0.0031]
	GPC	−0.0083	−0.0075	[−0.0208, +0.0059]
ToN-IoT (campaign)	SVC	−0.0172	−0.0148	[−0.0268, −0.0029]
	GPC	−0.0046	−0.0090	[−0.0190, +0.0010]
ToN-IoT (constructed)	SVC	−0.0016	−0.0011	[−0.0049, +0.0028]
	GPC	+0.0313	+0.0109	[+0.0017, +0.0202]
Dataset-equal mean, SVC		−0.0043	−0.0050	[−0.0060, −0.0040]
Dataset-equal mean, GPC		+0.0043	−0.0019	[−0.0028, +0.0003]

zero. The full-pool extended-family sensitivity is similar in direction and scale (dataset-equal means −0.004 for both classifiers), so the conclusion does not depend on giving the classical family its larger pool.

Rank matching does not reveal a residual quantum advantage.

We pair each quantum configuration with the classical configuration of nearest training effective rank, within the same run and dimension, and compare their OOD accuracy. Figure 4 reports the outcome: in all eight scenario-groups the median rank-matched delta is negative (quantum minus classical, between −0.001 and −0.009), and the quantum configuration is the more accurate of the pair in only 22–37% of matches. Within

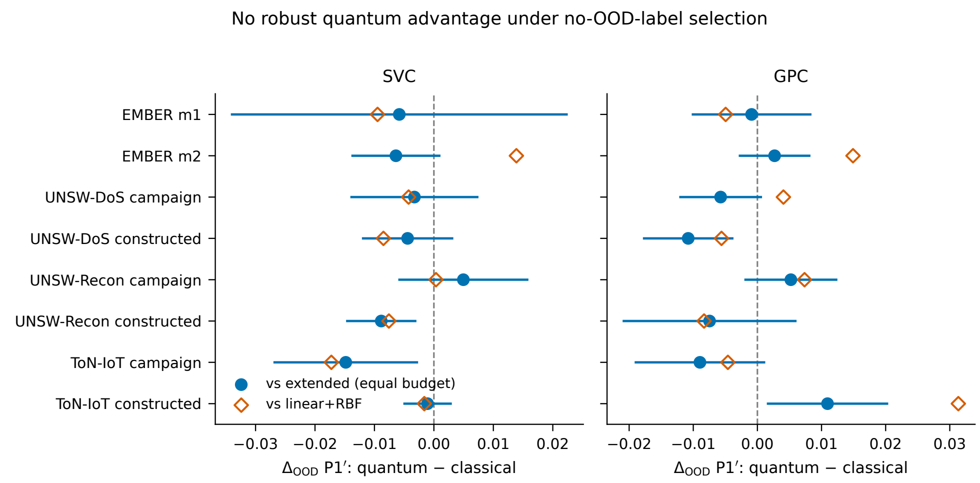

Fig. 3 No robust quantum advantage under no-OOD-label selection. Per-scenario-group P1′ OOD deltas (quantum minus classical), for SVC (left) and the GP classifier (right). Filled circles compare against the equal-budget extended classical family with conditional pipeline-realization intervals; open diamonds compare against the customary linear+RBF reference. All eight scenario-groups are shown.

the overlap covered by this observational matching, classical kernels are therefore at least as accurate at comparable effective rank.

2.5 Protocol sensitivity replicates on real-world domain shifts

We next applied the prospectively frozen protocol to College Scorecard, diabetes readmission, and ACS income, preserving the published TableShift domain splits and withholding all OOD labels from the controlled selector. The complete external grid contains 30 task–size–seed units and 37,800 split-level results; every prespecified unit passed the independent audit. Figure 5 shows all task–size effects and conditional seed-cluster intervals.

The primary SVC result satisfies the frozen criterion for *replicated protocol sensitivity*. The task-equal controlled effect is -0.0119 (conditional 95% interval $[-0.0205, -0.0033]$), whereas the weak-reference, fixed-C, same-test-oracle effect is $+0.0089$ ($[+0.0047, +0.0137]$). Their contraction is $+0.0208$ ($[+0.0113, +0.0307]$), with a leave-one-task-out range of $[+0.0116, +0.0260]$. No task has a positive controlled

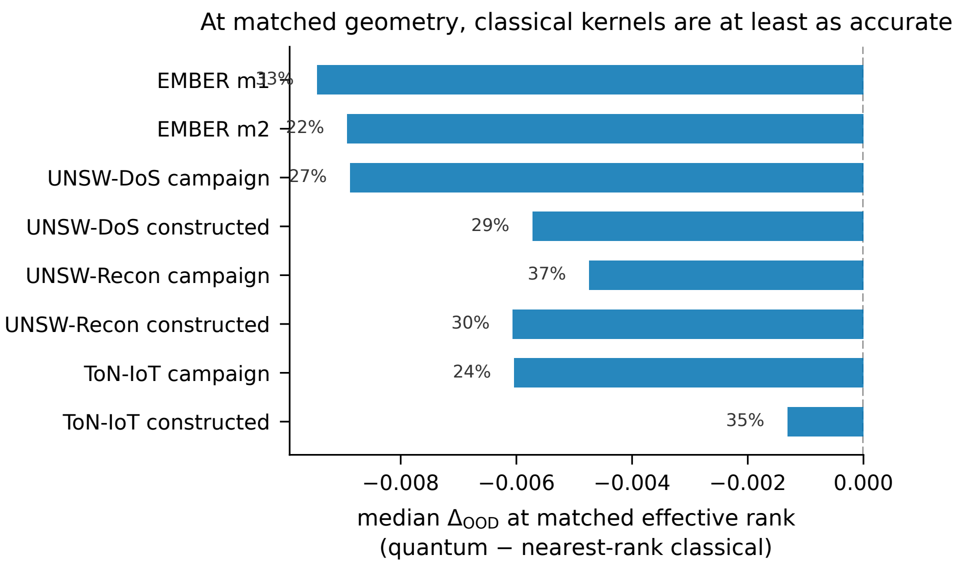

Fig. 4 At matched effective rank, classical kernels are at least as accurate. For each quantum configuration we find the classical configuration of nearest training effective rank (within run and dimension) and take the OOD balanced-accuracy difference; bars are per-scenario-group medians, annotated with the fraction of matched pairs in which the quantum configuration is the more accurate.

SVC effect whose interval excludes zero in both sizes. Size-equal controlled effects are -0.0160 for College Scorecard, -0.0066 for diabetes readmission, and -0.0131 for ACS income.

The secondary GP classifier shows a larger contraction of $+0.0324$ ($[+0.0223, +0.0427]$), but its task-equal controlled effect is $+0.0077$ with an interval spanning zero ($[-0.0015, +0.0183]$). College Scorecard leans quantum under this classifier, diabetes leans classical, and ACS is slightly positive. This classifier dependence limits the conclusion appropriately: the external study replicates the sensitivity of the family verdict to evaluation design, not universal classical dominance.

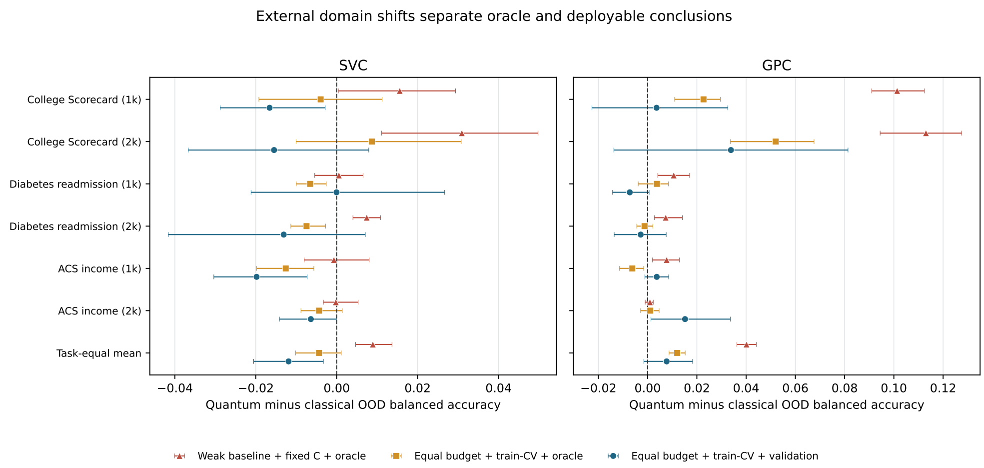

Fig. 5 External domain shifts separate optimistic and deployable conclusions. Effects are quantum minus classical OOD balanced accuracy for the weak-reference fixed-regularization same-test oracle, the equal-budget train-tuned same-test oracle, and the primary equal-budget train-tuned validation selector. Rows show both nested sizes for all three fixed tasks plus the task-equal mean. Intervals are conditional on the five seed clusters; no population p-value is reported.

2.6 Geometry is associated with, but does not determine, robustness

The two families are close because both reach similar kernel geometry, and geometry—not quantumness—is what carries information about behaviour under shift. That information, however, is partial: it is a regime-dependent association, not the single governing law that optimistic accounts (our own earlier report included) proposed.

The geometry–OOD link is a regime-dependent association, not a law.

Two descriptors of the training and OOD Gram matrices carry information about which kernels generalize: the effective rank of the training kernel and the centered kernel-target alignment on the OOD split. But their association with OOD accuracy is regime-dependent, not a single mechanism. Within scenario-groups (Figure 6 and Supplementary Table 3), the median Spearman correlation between effective rank and OOD accuracy ranges from +0.80 (EMBER $m2$) down to −0.15 (UNSW-NB15 reconnaissance under the constructed shift); partial correlations that control for family

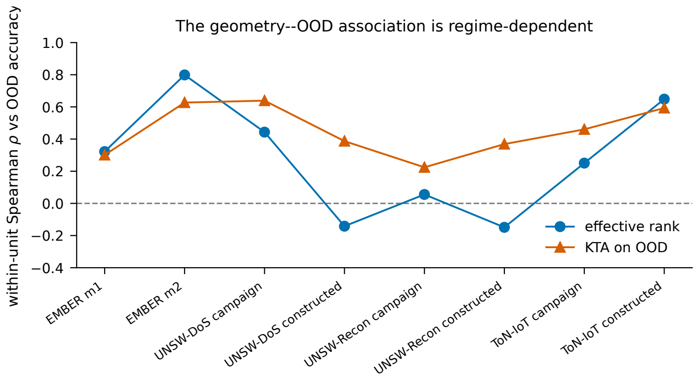

Fig. 6 The geometry–OOD association is regime-dependent. Median within-unit Spearman correlation between each geometry descriptor and OOD balanced accuracy, per scenario-group (SVC). Effective rank orders accuracy strongly in some regimes (EMBER $m2$, ToN-IoT $m2$) and weakly or negatively in others (the UNSW constructed-shift scenarios); OOD alignment is more consistently positive but never uniform. No single descriptor governs all regimes.

label and length scale track the raw values closely. OOD alignment is more consistently positive (+0.22 to +0.64) but likewise varies by regime. A geometry model fit on three datasets and used to rank the held-out dataset's kernels achieves only Spearman 0.23–0.40. We therefore describe a genuine but partial association, not a causal mechanism or deployable predictor.

2.7 Finite-shot estimation: the geometry is a statevector idealization

All quantum kernels above are computed by exact statevector simulation, which is the right idealization for a geometry study but not what a device would return. To probe how the picture degrades away from that idealization we apply a *finite-shot fidelity-estimation perturbation model*: each fidelity entry is replaced by a binomial estimate from s shots, train blocks are symmetrized, unit-diagonal, clipped to $[0, 1]$, and projected to the positive-semidefinite cone (Section 4). This is not a hardware simulation—no device noise, mitigation, or transpilation is modelled—but it isolates the statistical cost of estimating a fidelity from finite measurements. Table 2 reports the effect on the $q1000$ subset. OOD alignment and OOD accuracy are robust: median

Table 2 Finite-shot fidelity-estimation perturbation (EMBER+network $q1000$ subset, SVC). "OOD acc. dev." is the median absolute deviation of OOD balanced accuracy from the statevector-exact value; "KTA dev." the same for OOD alignment; "rank ratio" the median ratio of perturbed to exact effective rank; "sel. agree" the fraction of runs whose ID-validation-selected configuration is unchanged; "regret" the mean OOD accuracy lost by deploying the perturbed selection.

Shots	OOD acc. dev.	KTA dev.	rank ratio	sel. agree	regret
128	0.0327	0.0004	2.12	7%	+0.0007
512	0.0273	0.0001	1.56	10%	+0.0017
2048	0.0240	0.0001	1.29	19%	−0.0025
8192	0.0181	0.0000	1.15	21%	−0.0001

$\mathrm{KTA}_{\mathrm{OOD}}$ deviates by at most 0.0004 from the exact value even at 128 shots, and OOD balanced accuracy by 0.018–0.033. Effective rank is not: finite sampling inflates it by a median factor of 2.1 at 128 shots, falling to 1.15 only at 8192 shots, because sampling noise adds spurious spectral mass. The within-family configuration selected by ID-validation changes identity under perturbation (agreement 7–21%), but the accuracy cost of that instability is negligible (mean regret ≤ 0.003), so the flatness of the family optimum—already visible in the exact results—persists. The practical reading: the alignment description we can measure transfers to finite shots, but the effective-rank description, and any story that leans on it, is a statevector-exact quantity that would need many shots to reproduce.

2.8 Predictive uncertainty under shift

The Laplace GP probabilities provide a secondary uncertainty check (Supplementary Figure 1). For the honestly selected candidate, predictive entropy rises from ID to OOD for both families in every scenario-group, while expected calibration error changes moderately and comparably. Because neither family has an accuracy advantage, the fidelity kernels show no calibration benefit unavailable to classical kernels. We do not interpret rising entropy alone as evidence of calibration.

3 Discussion

The study's central interpretive move is to replace the question "do quantum kernels beat classical kernels under shift?" with "which properties of a kernel are associated with its behaviour under shift, and does either family have exclusive access to them?". Read through that lens, the results are clear and negative. The apparent advantage over linear and RBF baselines does not survive the combined correction of oracle selection, fixed regularization, and unequal candidate families. It does not reappear against the equal-budget extended family or at matched effective rank. The geometry descriptors carry information as regime-dependent associations rather than a governing law, and they are shared: a well-tuned heavy-tailed classical kernel occupies the same observed geometric regime as the fidelity maps. On present evidence there is no robust out-of-distribution advantage for fidelity kernels in the low-qubit settings studied, and the only residual—a modest in-distribution separability edge under the probabilistic classifier—is small, classifier-dependent, and itself shrinks under no-OOD-label selection.

From the perspective of the distribution-shift literature, this reinforces a general lesson: conclusions about robustness are only as credible as the protocol used to obtain them Quiñonero-Candela et al. (2008); Gulrajani and Lopez-Paz (2021); Koh et al. (2021); Yao et al. (2022). Our results add a QML-specific corollary. The strength of the classical baseline, whether its regularization is tuned, whether model selection may consult the target labels, and the size of each family's candidate budget are not implementation details: each can move the quantum-versus-classical verdict, and together they turn an apparent +0.037 advantage into a null. A demonstration that fixes any of them in the quantum family's favour, reported alone, would mislead—and much of the optimistic quantum-kernel literature fixes several at once.

The external validation shows that this is not only a peculiarity of constructed security shifts. Under a protocol frozen before result inspection, three education, health, and socioeconomic tasks reproduce a positive protocol contraction and a negative controlled SVC effect. The result is stable to omitting any one task, yet heterogeneous by task and classifier: the GP endpoint retains a small positive task-equal lean whose interval crosses zero. The strongest defensible conclusion is therefore about evaluation sensitivity, not the universal ordering of kernel families.

In-distribution separability is the one axis on which the fidelity kernels retain any edge, and even it must be stated carefully. Under the oracle that selects and evaluates on the same in-distribution split, the quantum family separates the classes slightly better than linear+RBF in every EMBER setting; but under honest selection—choosing on a validation split and reporting on a disjoint test split—the edge falls to +0.003 under SVC (mixed in sign across groups) and +0.012 under the probabilistic classifier, and it does not translate into any out-of-distribution benefit. This ID–OOD dissociation is real but modest, and it cuts against, not for, an operational recommendation: a family that is marginally more separable in-distribution yet no more robust under shift offers nothing to a deployment whose whole difficulty is the shift. The pattern is consistent with the "accuracy on the line" literature Miller et al. (2021), in which ID and OOD accuracy usually move together; here they decouple weakly, and geometry—shared by both families—rather than quantumness marks the axis along which they do.

The geometry association connects several strands of literature. That length scale is decisive for kernel behaviour is known for both families Shaydulin and Wild (2022); Canatar et al. (2023); our contribution is the comparative consequence—once length scale is tuned symmetrically and regularization is tuned at all, the classical family reaches the same geometric regime, so asymmetric tuning does not merely bias quantum-versus-classical comparisons, it manufactures the apparent advantage. The geometric difference of Huang et al. Huang et al. (2021) does not predict which family wins a scenario, consistent with its definition as a bound on the *potential* for a separation rather than its sign; a large geometric difference is compatible with equal or worse OOD accuracy, as we observe. That effective rank and OOD alignment are associated with performance is a statement about kernels in general, not about quantum kernels—which is precisely why the fidelity maps hold no privileged position: they are one costly route into a geometric regime that a short-length-scale classical kernel reaches directly.

The high-rank finding must also be reconciled with two spectral results that superficially point the other way. Kübler et al. argue that quantum kernels generalize in-distribution only when the relevant spectral mass is concentrated in few components Kübler et al. (2021), and Thanasilp et al. show that fidelity kernels concentrate exponentially—toward effectively rank-one, uninformative Gram matrices—as qubit counts grow Thanasilp et al. (2024). There is no contradiction, but there is a genuine tension that our results locate precisely. Both results concern the asymptotic or i.i.d. regime; our effective ranks (roughly 10–150 on training sets of 10^3–4×10^3 samples at 4–12 qubits) sit far below the concentration regime, so no contradiction arises. Our finite-shot analysis, however, adds an empirical caveat these theoretical results anticipate: effective rank measured from finitely estimated fidelities inflates substantially (Section 2.7), so the exact effective ranks we report are a statevector idealization, and any account that leans on effective rank as the operative quantity would need many shots to reproduce even at the modest qubit counts studied here.

We emphasize what the geometry analysis does not deliver. It is a post-hoc association, not a deployable selection rule: choosing a configuration by training-set geometry alone does not reliably identify the better kernel, and OOD alignment requires OOD labels. Because the two families are statistically indistinguishable under honest selection, the more useful open problem is no longer "which family wins" but whether any label-free, train-only signal can select a robust kernel at all—a question that concerns classical and quantum kernels equally.

From a practical perspective, the paper suggests three takeaways for security ML. The first is substantive: on the evidence here, fidelity-based quantum kernels do not offer a robustness advantage under distribution shift over well-tuned classical kernels, while their pairwise overlap estimation introduces circuit and shot costs absent from direct evaluation of the classical baselines. A median-heuristic Laplacian kernel, especially one with a shortened length scale and its regularization tuned, is a strong and inexpensive robustness baseline, and it matches the fidelity maps under no-OOD-label selection on both modalities. Practitioners weighing a quantum-kernel pipeline for OOD-sensitive security detection should, on this evidence, prefer a tuned classical kernel unless a specific regime demonstrably favours the quantum construction after the same controls.

The second is methodological. Independently of the preferred kernel family, the protocol used here provides a reusable blueprint for robust evaluation in security settings:

1. keep the classifier and preprocessing fixed, and swap only the kernel;
2. define explicit OOD splits tied to interpretable shift mechanisms, including at least one non-constructed shift mechanism where the data permit;
3. give every family the same freedom—the same candidate budget, symmetric length-scale tuning, and per-configuration regularization tuned on training data alone;
4. select the deployed configuration without out-of-distribution labels, and keep an in-distribution test split disjoint from the validation split used for selection;
5. repeat the pipeline, report variability as conditional intervals, and treat correlated benchmarks as fixed case studies rather than pooling them into a single population p-value.

Security ML frequently suffers from comparisons in which model class, tuning effort, and evaluation conditions change simultaneously; the way the verdict of Sections 2.2–2.4 moves from a large apparent advantage to a null as the protocol is made fair is a concrete demonstration of how much the conclusion depends on getting this right.

The third concerns diagnostics: the geometry descriptors are cheap to compute from Gram matrices that the pipeline already builds, and the alignment-survival analysis can be run post hoc on any deployed kernel system with a labeled drift sample. Practitioners should nonetheless resist reading these results as a blanket recommendation to replace classical kernels; the responsible recommendation is to test the candidate families under the expected shift regime, using exactly the kind of protocol described here.

Several limitations define the scope of our conclusions.

Our shift operationalization covers three reproducible mechanisms—within-class sparsity-tail extremes, train-geometry-tail extremes, and a held-out attack-campaign shift on network traffic (an unseen attack campaign plus a capture-partition change, not a timestamped temporal split)—but not the full space of real-world drift processes. Explicit temporal splits on timestamped malware, family-based drift, and adversarially induced shift remain outside the study's scope, and our conclusions should be re-tested under them rather than assumed.

Kernel and encoding coverage also remains finite. The classical family now spans seven kernels plus a bandwidth sweep, and both families receive symmetric tuning freedom, but neither candidate set is exhaustive: the quantum side is restricted to four commonly used fidelity feature maps at low qubit counts, and trained or task-adapted encodings, projected kernels, and deeper maps are not evaluated. The mechanism gives a concrete prediction for such extensions—what should matter is their Gram-matrix geometry, not their provenance—which future work can falsify. At substantially larger qubit counts, exponential kernel concentration Thanasilp et al. (2024) would erode the alignment-preserving rank the mechanism rewards, so our conclusions should not be extrapolated beyond the low-qubit regime studied here.

Selection is necessarily over finite candidate sets. P1′ (ID-validation selection), P2 (cross-realization OOD-supervised selection), and P3 (same-test OOD oracle) are selection procedures over finite candidate sets. The primary comparison matches candidate budgets, and the full-pool and budget curves provide sensitivities, but residual dependence on candidate definitions and search granularity remains. Fully nested selection over additional independent benchmark pools would strengthen the analysis further.

The quantum kernels are simulated rather than executed on hardware. The study therefore characterizes the geometry of fidelity kernels, not the behavior of noisy devices: finite sampling of kernel entries, hardware noise, and error mitigation would all perturb the Gram matrices Wang et al. (2021), and exponential concentration implies that with polynomially many shots the very spectral richness our mechanism rewards becomes expensive to resolve at larger scales Thanasilp et al. (2024); the effect of either on alignment survival is unknown and worth measuring. We make no claim about hardware feasibility, resource advantage, or quantum speedup—the only rigorously established quantum-classical learning separation concerns a specifically constructed problem, not natural data Liu et al. (2021). At the dimensions studied, the fidelity kernels are also classically simulable Schreiber et al. (2023); Jerbi et al. (2024), which is precisely what makes them available as practical kernel constructions today.

Dataset scope and feature caveats provide a final boundary. Four benchmark scenarios across two modalities is a substantial widening relative to single-dataset studies, but security is broader still: dynamic malware analysis and encrypted traffic remain untested. Within the network scenarios, the feature sets inherited from the public benchmarks retain port numbers and protocol indicators, which are legitimate but known shortcut-prone features in NIDS benchmarks Engelen et al. (2021);

because every kernel receives identical features, this affects absolute accuracies, not the family comparisons. Finally, the mechanism's effective-rank component is itself regime-dependent (strong on EMBER and ToN-IoT, weak on UNSW-Reconnaissance), a gradation we report rather than smooth over.

In light of these limitations, the appropriate interpretation of the paper is deliberately conservative. We do not claim universal superiority, complexity-theoretic quantum advantage, or hardware relevance—nor do we claim to have refuted quantum kernels in general. We claim three narrower things, supported by conditional effect estimates rather than a population p-value: that the apparent out-of-distribution advantage of fidelity kernels over customary baselines disappears when same-test oracle selection and untuned regularization are removed; that under equal candidate budgets and no-OOD-label selection no robust family-level advantage remains against either the customary or the extended classical reference, across eight fixed scenario-groups and two classifiers; and that the geometric descriptors associated with out-of-distribution behaviour occur in both families and do not identify kernel provenance as the source of robustness. This view is consistent with recent calls for careful benchmarking in QML Bowles et al. (2024); Schnabel and Roth (2025); Kakavand et al. (2026) and is more informative than either an unqualified advantage claim or an unqualified dismissal: it identifies the evaluation choices that generate the optimistic result and shows how the conclusion changes when they are controlled.

This paper asked why kernel families behave differently under distribution shift, using the quantum-versus-classical comparison as both motivation and stress test. A controlled kernel-swap protocol—classifier, preprocessing, splits, selection logic, and tuning freedom held fixed while only the kernel varies—was instantiated on four security scenarios spanning two modalities and three shift mechanisms, two classifier families including a probabilistic one, and 72 principal settings with fifteen pipeline realizations. Effects are reported as fixed-case summaries with conditional pipeline-realization intervals, not as population significance tests.

The empirical arc has three movements. Under same-test oracle selection and fixed regularization, fidelity-based quantum kernels appear to beat the customary linear and RBF baselines under shift. That result does not survive the controlled comparison: once each configuration is tuned on training data, selection uses no OOD labels, and both extended families search an equal budget of kernel geometries, the dataset-equal-weighted OOD difference is -0.005 under the primary SVC analysis and -0.002 under the GP classifier. The primary comparison is classical-favoured or tied in seven of eight scenario-groups. At matched effective rank the classical kernels are at least as accurate everywhere. Effective rank and OOD alignment are associated with performance in a regime-dependent way, are shared by both families, and, a finite-shot analysis shows, do not transfer unchanged from exact statevectors to finite measurement.

The contribution is therefore threefold. Empirically, the study provides a controlled, variability-aware comparison of quantum and classical kernels under shift whose central finding is a clean negative obtained under fair conditions: no robust family-level advantage survives budget matching, regularization tuning, and honest selection. Methodologically, it isolates the specific evaluation choices—oracle selection, fixed regularization, mismatched budgets, and pooled significance testing over correlated settings—that manufacture apparent quantum-kernel advantages, and it leaves behind a reproducible protocol that removes each. Analytically, it shows that the geometric quantities often invoked to explain such advantages describe both families equally and do not transfer unchanged to finite-shot estimation.

The most useful open problem this leaves is not "which family wins" but whether any label-free, train-only signal can select a robust kernel under shift at all; the security community's label-free drift detectors Yang et al. (2021) and the theory of task–model alignment Bordelon et al. (2020); Canatar et al. (2021); Ma et al. (2023) are natural starting points. It is a question about kernels in general, and answering it would benefit classical and quantum methods alike—which is, in the end, the point: the productive question is not whether one family dominates, but which similarity geometry aligns with the shift regime one actually faces, and how to find it without peeking.

4 Methods

4.1 Problem setting

We study binary security classification—malware versus benign software, and attack versus benign traffic—under distribution shift. Let D_{ID} denote the in-distribution data-generating process and D_{OOD} an out-of-distribution process induced by a constructed stress test or a held-out attack campaign. Given training data $T \sim D_{\mathrm{ID}}$, we evaluate on an ID test set S_{ID} and an OOD test set S_{OOD}.

To isolate the kernel, classifier, preprocessing, optimization protocol, and split design are fixed. Classical and quantum Gram matrices are therefore consumed by the same downstream learner on the same samples.

4.2 Datasets and split construction

The four scenarios span static malware (EMBER) Anderson and Roth (2018) and network flows: UNSW-NB15 DoS, UNSW-NB15 Reconnaissance Moustafa and Slay (2015), and ToN-IoT Scanning Alsaedi et al. (2020). Every setting has shared T, S_{ID}, and S_{OOD} samples with identical class balance across kernel families.

EMBER uses two label-conditional tail stress tests. The $m1$ split holds out within-class extremes of feature-block sparsity; $m2$ holds out within-class extremes of distance to train-fitted class centroids. Network scenarios use an analogous train-centroid construction and a held-out attack-campaign regime. In the latter, training and ID pools contain benign traffic plus non-target attacks, whereas OOD contains benign traffic plus the target campaign; UNSW pools additionally differ in capture partition. These are reproducible stress tests, not timestamped temporal splits or certified covariate shifts. Exact algorithms, feature exclusions, leakage checks, and pool definitions are in the Supplementary Note “Dataset and shift construction.”

4.3 Controlled kernel-swap protocol

Figure 1 summarizes the controlled pipeline. Both SVC and a binary Gaussian-process classifier with a logistic-link Laplace approximation consume unit-diagonal precomputed Gram matrices Cortes and Vapnik (1995); Rasmussen and Williams (2006). Comparisons are always within a classifier. The GP classifier supplies approximate probabilities for uncertainty diagnostics; no post-hoc calibration is applied.

Training-fitted MaxAbs scaling and TruncatedSVD produce shared representations at $d \in \{4, 6, 8, 10, 12\}$. Quantum inputs are then mapped linearly to $[0, \pi]$. Dimension is an explicit candidate axis.

4.4 Kernel families

The customary classical reference contains linear and RBF kernels. The extended reference adds degree-2 and degree-3 polynomial, Laplacian, and Matérn-3/2 and 5/2 kernels. RBF, Laplacian, and Matérn bandwidths sweep factors $\{0.1, 0.3, 1, 3, 10\}$ around training-only scale or median-distance heuristics. The quantum family uses fidelity

$$K_Q(x, x') = |\langle \phi_Q(x), \phi_Q(x') \rangle|^2$$

for two ZZ depths, PauliXZ, and Z feature maps Havlíček et al. (2019); Schuld and Killoran (2019). Every map sweeps angle scales $\{0.5, 1, 2\}$. Classical bandwidth and quantum angle-scale freedom are symmetric in purpose Shaydulin and Wild (2022); Canatar et al. (2023). Exact formulas and grids appear in the Supplementary Note “Controlled kernel-swap protocol.”

4.5 Selection, candidate budgets, and inference

For each geometry, SVC regularization $C \in \{0.01, 0.1, 1, 10, 100\}$ is selected by five-fold stratified cross-validation on the training Gram matrix. The primary P1$'$ protocol divides the ID holdout into disjoint validation and test halves. Within each family, the candidate with the highest ID-validation balanced accuracy is deployed on ID-test and OOD-test; OOD labels never enter this choice. Same-test OOD-oracle and cross-realization OOD-supervised protocols are descriptive sensitivities.

Best-of-family selection favours a larger pool. The primary extended-family endpoint therefore uses 60 candidates per family where coverage is complete and 20 where it is reduced, generated by 5,000 classical-pool subsamples under three schemes. The full 115-candidate classical pool is secondary.

Every configuration has fifteen pipeline realizations (five split seeds by three model seeds). Two EMBER and six network scenario-groups are crossed with three master seeds and $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$ regimes $(1000, 500, 500)$, $(2000, 1000, 1000)$, and $(4000, 1800, 1800)$. Because scenario-groups share source pools and are not exchangeable, they are fixed case studies. Per group we average over the five split-seed clusters and report a conditional interval; across groups we use dataset-equal means and leave-one-dataset-out ranges. We attach no population p-value to eight fixed, correlated cases. The complete grid is in Supplementary Table 1 and resampling details in the Supplementary Note "Controlled kernel-swap protocol."

4.6 Specification curve

The specification curve is a descriptive multiverse over every logically valid endpoint for which the frozen legacy and v4 artifacts contain complete paired data. Its ten specifications cross, where available, legacy or v4 generation; same-test OOD-oracle, legacy ID-based, or disjoint ID-validation selection; fixed $C = 1$ or five-fold train-CV regularization for SVC; customary linear/RBF or extended classical references; and native or equal candidate budgets. The order S1–S10 was fixed before aggregation and is never sorted by effect size. Within each specification, pipeline realizations are first averaged within setting and then within scenario-group; groups are averaged within source dataset before the three datasets receive equal weight. All eight group effects are displayed, and no specification-level hypothesis test is performed. Exact definitions and machine-readable group- and dataset-level values are provided in the Supplementary Information and versioned result artifact.

4.7 Prospectively frozen external validation

Before acquiring or inspecting any external quantum–classical result, we froze the tasks, sampling, preprocessing, candidate pools, estimands, uncertainty procedure, and interpretation rule in a version-controlled protocol. The external source is TableShift, a benchmark of real-world distribution shifts in tabular binary classification Gardner et al. (2023). We fixed three public tasks from distinct domains and with distinct published split variables: College Scorecard (institution type), diabetes readmission (admission source), and ACS income (geographic region). TableShift's predefined `train`, `validation`, `id_test`, and `ood_test` roles are preserved; its `ood_validation` split is excluded from preprocessing, tuning, selection, and reporting.

Each task has two nested sample-size strata: $(n_{\mathrm{train}}, n_{\mathrm{val}}, n_{\mathrm{ID}}, n_{\mathrm{OOD}}) = (1000, 250, 250, 500)$ and $(2000, 500, 500, 1000)$. Five label-blind deterministic subsampling seeds form the conditional resampling clusters. Training-only column filtering, numeric median and categorical mode imputation, one-hot encoding, MaxAbs scaling, TruncatedSVD, standardization, and range mapping produce the same dimensions and kernel pools as in the security study. The primary deployed configuration is selected on `validation`; `ood_test` is touched only for final evaluation. The same-test OOD oracle is retained as a non-deployable diagnostic.

For each task, size, seed, and classifier, the 60-candidate quantum pool is compared with 5,000 equal-size, whole-kernel-block subsamples of the 115-candidate classical pool. The external primary effect is the selected quantum OOD accuracy minus the expected selected classical OOD accuracy over these budget-matched subsamples. The protocol-contraction diagnostic subtracts this controlled effect from the weak-reference, fixed-C, same-test-oracle effect. We average seeds within task and size, give the two nested sizes equal weight within task, and give the three fixed tasks equal weight. Conditional 95% intervals use 9,999 resamples of seed clusters within task; no population p-value is attached to the three deliberately selected tasks. Full acquisition provenance, integrity gates, and the frozen outcome-classification rule are reported in the Supplementary Information.

4.8 Metrics and geometry analysis

OOD balanced accuracy is primary. Family effects are quantum minus classical after selection. ID balanced accuracy and ID-to-OOD degradation are secondary. For the GP classifier we also record log loss, Brier score, 15-bin expected calibration error Guo et al. (2017), and predictive entropy.

From the classifier Gram matrices we compute effective rank Roy and Vetterli (2007), centered kernel-target alignment Cristianini et al. (2001); Cortes et al. (2012), entry concentration, cross-split similarity, and regularized geometric difference Huang et al. (2021). We relate geometry to OOD accuracy within scenario-groups, use partial association to control family and length scale, test portability by leaving one dataset out, and pair quantum candidates with nearest-effective-rank classical candidates within run and dimension. These analyses are associational. Formulas and diagnostic definitions appear in the Supplementary Note “Metrics and geometry descriptors.”

4.9 Finite-shot perturbation

Exact fidelity entries are replaced by binomial estimates at $s \in \{128, 512, 2048, 8192\}$ shots. Training blocks are symmetrized, clipped, set to unit diagonal, and projected to the positive-semidefinite cone. The model isolates sampling error, not device noise. Projection diagnostics and interpretation are in the Supplementary Note "Finite-shot fidelity-estimation model."

4.10 Use of generative AI

OpenAI Codex was used during manuscript revision to assist with language editing, code inspection, and consistency checking. The authors reviewed and take responsibility for all analyses, claims, citations, and text; the system is not an author.

Declarations

Data availability. This study uses the public EMBER Anderson and Roth (2018), UNSW-NB15 Moustafa and Slay (2015), ToN-IoT Alsaedi et al. (2020), and TableShift Gardner et al. (2023) benchmarks, which remain subject to their original access conditions. Raw TableShift records are not redistributed. Derived split definitions, deterministic row-position hashes, shift constructions, experiment manifests, aggregated results, and summary tables are available in the Zenodo repository at https://doi.org/10.5281/zenodo.19147649.

Code availability. Code for constructing the splits, running the kernel experiments, reproducing the confirmatory analysis, and regenerating every table and figure is available at https://github.com/roberto-fernandez-barrios/kernel_shift_framework and in the versioned Zenodo archive above. The platform-independent Python 3.12 environment is pinned in `environment.lock.yml`; the code is released under BSD-3-Clause with no restriction on non-academic use.

Competing interests. The authors declare that they have no competing interests.

Authors' contributions. RFB: conceptualization, methodology, software, data curation, formal analysis, visualization, investigation, writing—original draft, writing—review and editing. IPL: supervision, validation, writing—review and editing. AGS: validation, project administration, writing—review and editing. PGB: supervision, resources, writing—review and editing. All authors read and approved the final manuscript.

Acknowledgements. The authors received no specific funding for the submitted work.

References

Abreu, D., Esteve Rothenberg, C., Abelém, A.: QML-IDS: Quantum machine learning intrusion detection system. In: 2024 IEEE Symposium on Computers and Communications (ISCC), pp. 1–6. IEEE, ??? (2024). https://doi.org/10.1109/ISCC61673.2024.10733655

Abreu, D., Moura, D., Esteve Rothenberg, C., Abelém, A.: QuantumNetSec: Quantum machine learning for network security. International Journal of Network Management **35**(4), 70018 (2025) https://doi.org/10.1002/nem.70018

Alsaedi, A., Moustafa, N., Tari, Z., Mahmood, A., Anwar, A.: TON_IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems. IEEE Access **8**, 165130–165150 (2020) https://doi.org/10.1109/ACCESS.2020.3022862

Arp, D., Quiring, E., Pendlebury, F., Warnecke, A., Pierazzi, F., Wressnegger, C., Cavallaro, L., Rieck, K.: Dos and don'ts of machine learning in computer security. In: 31st USENIX Security Symposium, pp. 3971–3988 (2022)

Anderson, H.S., Roth, P.: Ember: An open dataset for training static PE malware machine learning models. arXiv **abs/1804.04637** (2018)

Bowles, J., Ahmed, S., Schuld, M.: Better than classical? the subtle art of benchmarking quantum machine learning models. arXiv **abs/2403.07059** (2024) https://doi.org/10.48550/arXiv.2403.07059

Bordelon, B., Canatar, A., Pehlevan, C.: Spectrum dependent learning curves in kernel regression and wide neural networks. In: Proceedings of the 37th International Conference on Machine Learning (ICML). PMLR, vol. 119, pp. 1024–1034 (2020)

Bellante, A., Fioravanti, T., Carminati, M., Zanero, S., Luongo, A.: Evaluating the potential of quantum machine learning in cybersecurity: A case-study on PCA-based intrusion detection systems. Computers & Security **154**, 104341 (2025) https://doi.org/10.1016/j.cose.2025.104341

Canatar, A., Bordelon, B., Pehlevan, C.: Spectral bias and task-model alignment explain generalization in kernel regression and infinitely wide neural networks. Nature Communications **12**, 2914 (2021) https://doi.org/10.1038/s41467-021-23103-1

Cortes, C., Mohri, M., Rostamizadeh, A.: Algorithms for learning kernels based on centered alignment. Journal of Machine Learning Research **13**(28), 795–828 (2012)

Canatar, A., Peters, E., Pehlevan, C., Wild, S.M., Shaydulin, R.: Bandwidth enables generalization in quantum kernel models. Transactions on Machine Learning Research (2023)

Cristianini, N., Shawe-Taylor, J., Elisseeff, A., Kandola, J.: On kernel-target alignment. In: Advances in Neural Information Processing Systems (NeurIPS), vol. 14 (2001)

Cortes, C., Vapnik, V.: Support-vector networks. Machine Learning **20**(3), 273–297 (1995) https://doi.org/10.1007/BF00994018

Engelen, G., Rimmer, V., Joosen, W.: Troubleshooting an intrusion detection dataset: The CICIDS2017 case study. In: 2021 IEEE Security and Privacy Workshops (SPW), pp. 7–12. IEEE, ??? (2021). https://doi.org/10.1109/SPW53761.2021.00009

Gulrajani, I., Lopez-Paz, D.: In search of lost domain generalization. In: International Conference on Learning Representations (2021). https://openreview.net/forum?id=lQdXeXDoWtI

Gardner, J., Popovic, Z., Schmidt, L.: Benchmarking distribution shift in tabular data with TableShift. In: Advances in Neural Information Processing Systems, vol. 36 (2023). https://doi.org/10.52202/075280-2324

Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: Proceedings of the 34th International Conference on Machine Learning (ICML). PMLR, vol. 70, pp. 1321–1330 (2017)

Huang, H.-Y., Broughton, M., Mohseni, M., Babbush, R., Boixo, S., Neven, H., McClean, J.R.: Power of data in quantum machine learning. Nature Communications **12**, 2631 (2021) https://doi.org/10.1038/s41467-021-22539-9

Havlíček, V., Córcoles, A.D., Temme, K., Harrow, A.W., Kandala, A., Chow, J.M., Gambetta, J.M.: Supervised learning with quantum-enhanced feature spaces. Nature **567**, 209–212 (2019) https://doi.org/10.1038/s41586-019-0980-2

Herrero Gómez, P., Jimeno Morenilla, A., Muñoz-Hernández, D., Mora Mora, H.: Spectral Phase Encoding for Quantum Kernel Methods. arXiv:2602.19644 (2026)

Jerbi, S., Gyurik, C., Marshall, S.C., Molteni, R., Dunjko, V.: Shadows of quantum machine learning. Nature Communications **15**, 5676 (2024) https://doi.org/10.1038/s41467-024-49877-8

Kübler, J.M., Buchholz, S., Schölkopf, B.: The inductive bias of quantum kernels. In: Advances in Neural Information Processing Systems (NeurIPS), vol. 34, pp. 12661–12673 (2021)

Koh, P.W., Sagawa, S., Marklund, H., Xie, S.M., Zhang, M., Balsubramani, A., Hu, W., Yasunaga, M., Phillips, R.L., Gao, I., Lee, T., David, E., Stavness, I., Guo, W., Earnshaw, B., Haque, I., Beery, S.M., Leskovec, J., Kundaje, A., Pierson, E., Levine, S., Finn, C., Liang, P.: Wilds: A benchmark of in-the-wild distribution shifts. In: Proceedings of the 38th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 139, pp. 5637–5664. PMLR, ??? (2021). https://proceedings.mlr.press/v139/koh21a.html

Kakavand, S., Strohmeyer, C., Schlotter, M.: Benchmarking quantum kernel support vector machines against classical baselines on tabular data: A rigorous empirical study with hardware validation. arXiv **abs/2604.18837** (2026) https://doi.org/10.48550/arXiv.2604.18837

Liu, Y., Arunachalam, S., Temme, K.: A rigorous and robust quantum speed-up in supervised machine learning. Nature Physics **17**, 1013–1017 (2021) https://doi.org/10.1038/s41567-021-01287-z

Ma, C., Pathak, R., Wainwright, M.J.: Optimally tackling covariate shift in RKHS-based nonparametric regression. The Annals of Statistics **51**(2), 738–761 (2023) https://doi.org/10.1214/23-AOS2268

Moustafa, N., Slay, J.: UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set),. In: 2015 Military Communications and Information Systems Conference (MilCIS), pp. 1–6 (2015). https://doi.org/10.1109/MilCIS.2015.7348942

Miller, J.P., Taori, R., Raghunathan, A., Sagawa, S., Koh, P.W., Shankar, V., Liang, P., Carmon, Y., Schmidt, L.: Accuracy on the line: On the strong correlation between out-of-distribution and in-distribution generalization. In: Proceedings of the 38th International Conference on Machine Learning (ICML). PMLR, vol. 139, pp. 7721–7735 (2021)

Pendlebury, F., Pierazzi, F., Jordaney, R., Kinder, J., Cavallaro, L.: TESSERACT: Eliminating experimental bias in malware classification across space and time. In: 28th USENIX Security Symposium, pp. 729–746 (2019)

Quiñonero-Candela, J., Sugiyama, M., Schwaighofer, A., Lawrence, N.D. (eds.): Dataset Shift in Machine Learning. MIT Press, Cambridge, MA (2008). https://doi.org/10.7551/mitpress/9780262170055.001.0001

Roy, O., Vetterli, M.: The effective rank: A measure of effective dimensionality. In: 2007 15th European Signal Processing Conference (EUSIPCO), pp. 606–610 (2007)

Rasmussen, C.E., Williams, C.K.I.: Gaussian Processes for Machine Learning. MIT Press, Cambridge, MA (2006). https://doi.org/10.7551/mitpress/3206.001.0001

Schuld, M.: Supervised quantum machine learning models are kernel methods. arXiv **abs/2101.11020** (2021) https://doi.org/10.48550/arXiv.2101.11020

Schreiber, F.J., Eisert, J., Meyer, J.J.: Classical surrogates for quantum learning models. Physical Review Letters **131**, 100803 (2023) https://doi.org/10.1103/PhysRevLett.131.100803

Schuld, M., Killoran, N.: Quantum machine learning in feature hilbert spaces. Physical Review Letters **122**(4), 040504 (2019) https://doi.org/10.1103/PhysRevLett.122.040504

Schnabel, J., Roth, M.: Quantum kernel methods under scrutiny: A benchmarking study. Quantum Machine Intelligence **7**, 58 (2025) https://doi.org/10.1007/s42484-025-00273-5

Schuld, M., Sweke, R., Meyer, J.J.: Effect of data encoding on the expressive power of variational quantum-machine-learning models. Physical Review A **103**(3), 032430 (2021) https://doi.org/10.1103/PhysRevA.103.032430

Shaydulin, R., Wild, S.M.: Importance of kernel bandwidth in quantum machine learning. Physical Review A **106**(4), 042407 (2022) https://doi.org/10.1103/PhysRevA.106.042407

Thanasilp, S., Wang, S., Cerezo, M., Holmes, Z.: Exponential concentration in quantum kernel methods. Nature Communications **15**, 5200 (2024)

Wang, X., Du, Y., Luo, Y., Tao, D.: Towards understanding the power of quantum kernels in the NISQ era. Quantum **5**, 531 (2021) https://doi.org/10.22331/q-2021-08-30-531

Yao, H., Choi, C., Cao, B., Lee, Y., Koh, P.W., Finn, C.: Wild-time: A benchmark of in-the-wild distribution shift over time. In: Advances in Neural Information Processing Systems (2022). Datasets and Benchmarks Track

Yang, L., Guo, W., Hao, Q., Ciptadi, A., Ahmadzadeh, A., Xing, X., Wang, G.: CADE: Detecting and explaining concept drift samples for security applications. In: 30th USENIX Security Symposium, pp. 2327–2344 (2021)

Zhou, K., Liu, Z., Qiao, Y., Xiang, T., Loy, C.C.: Domain generalization: A survey. IEEE Transactions on Pattern Analysis and Machine Intelligence **45**(4), 4396–4415 (2023) https://doi.org/10.1109/TPAMI.2022.3195549